MTH2003 Examples Sheet 2 Solutions

1. (***a***) **Question** $y'' + 8y' + 15y = -3\,\mathrm{e}^{-4t}$, $y(0) = 6$, $y'(0) = -25$.
Solution

[i] The left-hand side is with constant coefficients, so to find the complementary solution, we substitute $y = \mathrm{e}^{mt}$. This gives the characteristic (a.k.a. *auxiliary*) equation $m^2 + 8m + 15 = 0$. This quadratic equation has roots $m = -3$ and $m = -5$. Therefore the linearly independent solutions are e^{-3t} and e^{-5t}, and the complementary solution is $\boxed{y_c = c_1\,\mathrm{e}^{-3t} + c_2\,\mathrm{e}^{-5t}}$.

[ii] The index in the exponential on the right-hand side has $\alpha = -4$, which is not one of the roots of the characteristic equation, hence $s = 0$ and we get the particular solution by trying $y = B\,\mathrm{e}^{-4t}$. On substituting this into the nonhomogeneous differential equation and cancelling by e^{-4t}, we get $16B+8(-4B)+15B = -3$ so that $B = 3$ and the particular solution is $\boxed{y_p = 3\,\mathrm{e}^{-4t}}$.

[iii] The General Solution is therefore $\boxed{y = c_1\,\mathrm{e}^{-3t} + c_2\,\mathrm{e}^{-5t} + 3\,\mathrm{e}^{-4t}}$.

[iv] From the initial conditions: $y(0) = 6 = c_1+c_2+3$ and $y'(0) = -25 = -3c_1-5c_2-12$, so that $c_1 + c_2 = 3$ and $3c_1 + 5c_2 = 13$ so that $c_1 = 1$ and $c_2 = 2$. The solution of the initial value problem is therefore $\boxed{y = \mathrm{e}^{-3t} + 2\,\mathrm{e}^{-5t} + 3\,\mathrm{e}^{-4t}}$.

(***b***) **Question** $y'' + 6y' + 8y = 16\,\mathrm{e}^{-2t}$.
Solution

[i] The characteristic equation is $m^2+6m+8 = 0$ and the roots are $m = -2$ and $m = -4$. Therefore the linearly independent solutions are e^{-2t} and e^{-4t}, and the complementary solution is then $\boxed{y_c = C_1\,\mathrm{e}^{-2t} + C_2\,\mathrm{e}^{-4t}}$.

[ii] The right-hand side is an exponential with index $\alpha = -2$ which *is* one of the roots of the characteristic equation, and we have simple (degree $s = 1$) resonance so we try $Bt\,\mathrm{e}^{-2t}$.

$$\frac{\mathrm{d}y}{\mathrm{d}t} = B\,\mathrm{e}^{-2t} - 2tB\,\mathrm{e}^{-2t}, \qquad \text{and} \qquad \frac{\mathrm{d}^2y}{\mathrm{d}t^2} = 4Bt\,\mathrm{e}^{-2t} - 4B\,\mathrm{e}^{-2t}.$$

Substituting into the ODE gives:

$$\mathrm{LHS} = -4B\,\mathrm{e}^{-2t} + 4Bt\,\mathrm{e}^{-2t} + 6\left(B\,\mathrm{e}^{-2t} - 2Bt\,\mathrm{e}^{-2t}\right) + 8Bt\,\mathrm{e}^{-2t} = \mathrm{RHS} = 16\,\mathrm{e}^{-2t}.$$

We see that the terms in $t\,\mathrm{e}^{-2t}$ cancel out. Collecting the other terms and cancelling through by e^{-2t} gives $2B = 16$ so that $B = 8$. The particular solution is therefore $\boxed{y_p = 8t\,\mathrm{e}^{-2t}}$.

[iii] The General Solution is $\boxed{y = (8t + C_1)\,\mathrm{e}^{-2t} + C_2\,\mathrm{e}^{-4t}}$

(***c***) **Question** $y'' + 4y' + 13y = 136\cos(5t)$, $y(0) = 0$, $y'(0) = 25$.
Solution

[i] The characteristic equation is $m^2 + 4m + 13 = 0$. This has the complex roots $m = -2 \pm 3\mathrm{i}$. The real form of the complementary solution is then

$$\boxed{y_c = \mathrm{e}^{-2t}\left(C_1\cos(3t) + C_2\sin(3t)\right)}.$$

[ii] The right hand side is a cosine with $\beta = 5$, $\alpha = 0$, and $\alpha + i\beta = 5i$ is different to the roots $m = -2 \pm 3\mathrm{i}$ of the characteristic equation, so $s = 0$ (no resonances). We can obtain the particular solution by trying $y = B\cos(5t) + C\sin(5t)$. Then $y' = -5B\sin(5t) + 5C\cos(5t)$, $y'' = -25B\cos(5t) - 25C\sin(5t)$ and substituting into the ODE gives, on collecting terms proportional to $\sin(5t)$ and $\cos(5t)$ on the LHS:

$$(-25C - 20B + 13C)\sin(5t) + (-25B + 20C + 13B)\cos(5t) = 136\cos(5t)$$

Comparing the coefficients of $\sin(5t)$ on the LHS and RHS gives $-12C - 20B = 0$ (i.e. $C = -5B/3$), while comparing the coefficient of $\cos(5t)$ gives $-12B + 20C = 136$ and hence $-12B - 100B/3 = 136$, i.e. $B = -3$, $C = 5$ and a particular solution is

$$\boxed{y_p = -3\cos(5t) + 5\sin(5t)}.$$

[iii] The complete General Solution is

$$\boxed{y = \mathrm{e}^{-2t}\left(C_1\cos(3t) + C_2\sin(3t)\right) - 3\cos(5t) + 5\sin(5t).}$$

[iv] We now use the initial conditions to determine C_1 and C_2. First, $y(0) = C_1 - 3 = 0$. Therefore $C_1 = 3$. Second, $y'(0) = 3C_2 - 2C_1 + 25 = 25$ and so $C_2 = 2$. So the solution of the initial value problem is

$$\boxed{y = \mathrm{e}^{-2t}\left(3\cos(3t) + 2\sin(3t)\right) - 3\cos(5t) + 5\sin(5t).}$$

(***d***) **Question** $y'' + 8y' + 25y = 125t^2 + 5t - 14$, $y(0) = 5$, $y'(0) = -11$.
Solution

[i] Characteristic equation: $m^2 + 8m + 25 = 0$, $m = -4 \pm 3\mathrm{i}$, and complementary solution:

$$\boxed{y_c = \mathrm{e}^{-4t}[c_1\cos(3t) + c_2\sin(3t)]}.$$

[ii] The right-hand side is a polynomial and 0 is not a root of the characteristic equation so $s = 0$. For the particular solution, we try $y = B_2t^2 + B_1t + B_0$. Substitution gives $2B_2 + 8[2B_2t + B_1] + 25[B_2t^2 + B_1t + B_0] = 125t^2 + 5t - 14$. Equating the coefficients of t^2 on both sides gives $25B_2 = 125$ so that $B_2 = 5$. Equating the coefficients of t on both sides gives $16B_2 + 25B_1 = 5$ so that $B_1 = -3$. Equating the constant terms on both sides gives $2B_2 + 8B_1 + 25B_0 = -14$, so that $B_0 = 0$. Hence the particular integral is

$$\boxed{y_p = 5t^2 - 3t}.$$

[iii] The general solution is then $\boxed{y = \mathrm{e}^{-4t}[c_1\cos(3t) + c_2\sin(3t)] + 5t^2 - 3t}$.

[iv] From the initial conditions: $y(0) = 5 = c_1$, so $y'(t) = -4\,\mathrm{e}^{-4t}[c_1\cos(3t) + c_2\sin(3t)] + \mathrm{e}^{-4t}[-3c_1\sin(3t) + 3c_2\cos(3t)] + 10t - 3$, and $y'(0) = -11 = -4c_1 + 3c_2 - 3$ so that $c_2 = 4$. The solution is therefore $\boxed{y = \mathrm{e}^{-4t}[5\cos(3t) + 4\sin(3t)] + 5t^2 - 3t}$.

(*e*) **Question** $y'' + 6y' + 8y = \mathrm{e}^{3t}\sin(2t)$.

Solution

[i] The left-hand side and therefore the complementary solution are the same as in Problem 1b, $\boxed{y_c = C_1\,\mathrm{e}^{-2t} + C_2\,\mathrm{e}^{-4t}}$.

[ii] First of all, the right-hand side is a product of an exponential with $\alpha = 3$ and a sine with $\beta = 2$ and $\alpha + i\beta = 3 + 2i$ is not a root of the characteristic polynomial and so $s = 0$. We try a particular solution of the form $y = \mathrm{e}^{3t}(B\cos(2t) + C\sin(2t))$. Differentiating (using the product rule) and collecting terms proportional to $\sin(2t)$ and $\cos(2t)$ gives $y' = \mathrm{e}^{3t}(3B + 2C)\cos(2t) + \mathrm{e}^{3t}(-2B + 3C)\sin(2t)$ and $y'' = \mathrm{e}^{3t}(5B + 12C)\cos(2t) + \mathrm{e}^{3t}(-12B + 5C)\sin(2t)$ and substituting into the ODE gives

$$\mathrm{e}^{3t}\left[(31B + 24C)\cos(2t) - (24B - 31C)\sin(2t)\right] = \mathrm{e}^{3t}\sin(2t)$$

Comparing the coefficients of $\cos(2t)$ on the LHS and RHS gives $B = -24C/31$, while comparing the coefficient of $\sin(2t)$ gives $-24B + 31C = 1$ hence $B = -24/1537$ and $C = 31/1537$ and a particular solution is

$$\boxed{y_p = \frac{\mathrm{e}^{3t}}{1537}(31\sin(2t) - 24\cos(2t))}$$

[iii] The general solution is $\boxed{y = \dfrac{\mathrm{e}^{3t}}{1537}\,(31\sin(2t) - 24\cos(2t)) + C_1\,\mathrm{e}^{-2t} + C_2\,\mathrm{e}^{-4t}}$

(*f*) **Question** $y'' - 5y' + 6y = 4t\,\mathrm{e}^{4t}$, $y(0) = 5$, $y'(0) = 11$.

Solution

[i] The complementary solution is obtained from $m^2 - 5m + 6 = 0$, $m \in \{2, 3\}$, $\boxed{y_c = c_1\,\mathrm{e}^{2t} + c_2\,\mathrm{e}^{3t}}$.

[ii] Particular solution: On the RHS, the exponential has $\alpha = 4$ and the polynomial has $k = 1$. Since 4 is not a root of the characteristic polynomial we try $(B_0 + B_1 t)\,\mathrm{e}^{4t}$. This gives: $(16B_0 + 16B_1 t + 8B_1)\,\mathrm{e}^{4t} - 5\,[4B_0 + 4B_1 t + B_1]\,\mathrm{e}^{4t} + 6\,[B_0 + B_1 t]\,\mathrm{e}^{4t} = 4t\,\mathrm{e}^{4t}$. Cancelling through by e^{4t} gives $2B_0 + 3B_1 + 2B_1 t = 4t$. Equating the coefficients of t on both sides gives $B_1 = 2$ and equating the constant terms gives $B_0 = -3$. So the particular solution is $\boxed{y_p = (-3 + 2t)\,\mathrm{e}^{4t}}$.

[iii] The General Solution is then $\boxed{y = c_1\,\mathrm{e}^{2t} + c_2\,\mathrm{e}^{3t} + (2t - 3)\,\mathrm{e}^{4t}}$.

[iv] From the initial conditions: $y(0) = 5 = c_1 + c_2 - 3$ and $y'(0) = 11 = 2c_1 + 3c_2 - 10$, we get $c_1 = 3$ and $c_2 = 5$. The complete solution is then $\boxed{y = 3\,\mathrm{e}^{2t} + 5\,\mathrm{e}^{3t} + (2t - 3)\,\mathrm{e}^{4t}}$.

2. (***a***) **Question** $t(t-2)\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2}+(2-t^2)\dfrac{\mathrm{d}y}{\mathrm{d}t}+2(t-1)y=0, \quad t\neq 0,2, \quad y_1=\mathrm{e}^t.$

Solution

[i] With $y=\mathrm{e}^t$ we have $\mathrm{d}y/\mathrm{d}t=\mathrm{e}^t$ and $\mathrm{d}^2y/\mathrm{d}t^2=\mathrm{e}^t$ and so substituting into the left-hand side (LHS) of the ODE gives $\mathrm{e}^t(t^2-2t+2-t^2+2t-2)=0$, so $y_1=\mathrm{e}^t$ is a solution of the homogeneous equation, as required.

[ii] We now substitute $y=v\,\mathrm{e}^t$ into the differential equation, which gives $t(t-2)\left[v''+2v'+v\right]\mathrm{e}^t+(2-t^2)\left[v'+v\right]\mathrm{e}^t+(2t-2)v\,\mathrm{e}^t=0$. Open the square brackets; the terms involving v cancel out leaving only the derivatives of v. Collecting terms proportional to v'' and v', and introducing $w=v'$, we have $t(t-2)\dfrac{\mathrm{d}w}{\mathrm{d}t}+(t^2-4t+2)w=0$, or in the standard form for a first-order linear ODE

$$\frac{\mathrm{d}w}{\mathrm{d}t}+\frac{t^2-4t+2}{t^2-2t}w=0.$$

We can separate the variables (noting the special solution $w\equiv 0$) to get

$$\int\frac{\mathrm{d}w}{w}=\ln|w|=-\int\frac{t^2-4t+2}{t^2-2t}\,\mathrm{d}t=\int\left(-1+\frac{2t-2}{t^2-2t}\right)\mathrm{d}t=-t+\ln|t^2-2t|+C.$$

Therefore $w=A(t^2-2t)\,\mathrm{e}^{-t}=\dfrac{\mathrm{d}v}{\mathrm{d}t}$ (covering the special solution by $A=0$). Now we integrate w to find

$$v=A\int(t^2-2t)\,\mathrm{e}^{-t}\,\mathrm{d}t$$

and we need to use integration by parts. We have

$$\begin{aligned}\int(t-2)\,\mathrm{e}^{-t}\,\mathrm{d}t &= -(t^2-2t)\,\mathrm{e}^{-t}+\int\mathrm{e}^{-t}(2t-2)dt\\ &= -(t^2-2t)\,\mathrm{e}^{-t}+\left(-(2t-2)\mathrm{e}^{-t}+\int 2\mathrm{e}^{-t}dt\right)\\ &= -t^2\,\mathrm{e}^{-t}+2\mathrm{e}^{-t}-2\mathrm{e}^{-t}+B \qquad (1)\end{aligned}$$

and hence

$$v=-t^2\,\mathrm{e}^{-t}+B.$$

By assumption $y_2=v\,\mathrm{e}^t$, we can take, for example, $B=0$ and $A=-1$ to get $\boxed{y_2=t^2}$.

[iii] If $y=t^2$, then $y'=2t$ and $y''=2$. Substituting this into the ODE, we have

$$t(t-2)\cdot 2+(2-t^2)\cdot 2t+(2t-2)\cdot t^2=2t^2-4t+4t-2t^3+2t^3-2t^2=0. \quad \blacksquare$$

[iv] We have $W[y_1,y_2]=\begin{vmatrix} y_1 & y_2\\ y_1' & y_2'\end{vmatrix}=\begin{vmatrix} \mathrm{e}^t & t^2\\ \mathrm{e}^t & 2t\end{vmatrix}=\mathrm{e}^t(2t-t^2)\neq 0$ because the conditions $t\neq 0$ and $t\neq 2$ are given. Hence y_1 and y_2 are linearly independent.

(***b***) **Question** $2t^2\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2}+3t\dfrac{\mathrm{d}y}{\mathrm{d}t}-y=0, \quad t>0, \quad y_1=1/t.$

Solution

[i] With $y=1/t$ we have $\mathrm{d}y/\mathrm{d}t=-1/t^2$ and $\mathrm{d}^2y/\mathrm{d}t^2=2/t^3$, and so substituting into the LHS of the ODE gives $4t^2/t^3-3t/t^2-1/t=0$, as required.

[ii] We now substitute $y = vy_1 = v/t$ into the differential equation, which gives

$$2t^2\left[\frac{1}{t}\frac{\mathrm{d}^2v}{\mathrm{d}t^2} - \frac{2}{t^2}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2}{t^3}v\right] + 3t\left[\frac{1}{t}\frac{\mathrm{d}v}{\mathrm{d}t} - \frac{1}{t^2}v\right] - \frac{1}{t}v = 0.$$

The terms involving v cancel out leaving only the derivatives of v. If we also write $\mathrm{d}v/\mathrm{d}t = w$, we obtain

$$2t\frac{\mathrm{d}w}{\mathrm{d}t} - w = 0.$$

Separation of variables gives $w \equiv 0$ or

$$\int \frac{\mathrm{d}w}{w} = \frac{1}{2}\int \frac{\mathrm{d}t}{t}$$

and hence

$$\ln|w| = \frac{1}{2}\ln(t) + A$$

(note that the problem is restricted to the interval $t > 0$). Therefore $w = Bt^{1/2}$, so that $v = B\int t^{1/2}\,\mathrm{d}t = Ct^{3/2} + D$. The solution for y is then

$$y = \frac{v}{t} = (Ct^{3/2} + D)\frac{1}{t} = Ct^{1/2} + Dt^{-1}.$$

Here A, B, C and D are arbitrary constants, and C and D in the final expression are two independent arbitrary constants giving the general solution. The second linearly independent solution, paired to $y_1 = 1/t$, could be taken to be $\boxed{y_2 = t^{1/2}}$.

[iii] If $y = t^{1/2}$, then $y' = \frac{1}{2}t^{-1/2}$ and $y'' = -\frac{1}{4}t^{-3/2}$. Substituting this into the LHS of the ODE gives

$$2t^2\left(-\frac{1}{4}t^{-3/2}\right) + 3t\left(\frac{1}{2}t^{-1/2}\right) - t^{1/2} = 0. \quad \blacksquare$$

[iv] We have $W[y_1, y_2] = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} t^{-1} & t^{1/2} \\ -t^{-2} & \frac{1}{2}t^{-1/2} \end{vmatrix} = \frac{3}{2}t^{-3/2} \neq 0$ for $t > 0$, hence y_1 and y_2 are linearly independent.

(***c***) **Question** $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} - 2\dfrac{\mathrm{d}y}{\mathrm{d}t} + y = 0, \quad y_1 = \mathrm{e}^t.$

Solution

[i] If $y = \mathrm{e}^t$ then $\mathrm{d}y/\mathrm{d}t = \mathrm{d}^2y/\mathrm{d}t^2 = \mathrm{e}^t$, so that the LHS of the ODE reads

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 2\frac{\mathrm{d}y}{\mathrm{d}t} + y = (1 - 2 + 1)\mathrm{e}^t = 0\,,$$

i.e. $y = \mathrm{e}^t$ is a solution.

[ii] We now substitute $y = v\mathrm{e}^t$. This gives

$$\left(\frac{\mathrm{d}^2v}{\mathrm{d}t^2} + 2\frac{\mathrm{d}v}{\mathrm{d}t} + v\right)\mathrm{e}^t - 2\left(\frac{\mathrm{d}v}{\mathrm{d}t} + v\right)\mathrm{e}^t + v\mathrm{e}^t = 0.$$

Majority of the terms cancel, leaving only

$$\frac{\mathrm{d}^2v}{\mathrm{d}t^2}\mathrm{e}^t = 0.$$

Substituting $w = v'$ and dividing through by e^t we find

$$\frac{\mathrm{d}w}{\mathrm{d}t} = 0.$$

The solution is an arbitrary constant, $w = A = \mathrm{d}v/\mathrm{d}t$. Integrating this gives $v = At + B$, or $y = v\mathrm{e}^t = (At + B)\,\mathrm{e}^t$. The first (given) solution was $y_1 = \mathrm{e}^t$. So the second solution can be chosen, for example $A = 1$ and $B = 0$, which gives $\boxed{y_2 = t\,\mathrm{e}^t}$.

[iii] If $y = t\mathrm{e}^t$, then $y' = (t+1)\mathrm{e}^t$ and $y'' = (t+2)\mathrm{e}^t$. Substituting this into the LHS of the ODE gives

$$(t + 2 - 2(t+1) + t)\,\mathrm{e}^t = 0.$$

[iv] We have $W[y_1, y_2] = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \mathrm{e}^t & t\mathrm{e}^t \\ \mathrm{e}^t & (t+1)\mathrm{e}^t \end{vmatrix} = \mathrm{e}^{2t} \neq 0$, hence y_1 and y_2 are linearly independent.

(***d***) **Question** $t^2y'' - 4ty' + 6y = 0, \quad t \neq 0, \quad y_1 = t^2.$

Answer

$$\boxed{y_2 = t^3.}$$

3. (***a***) **Question** $y'' + 6y' + 9y = \frac{1}{t^2}\,e^{-3t}, \quad t \neq 0.$

Solution The homogeneous equation $y'' + 6y' + 9y = 0$ has constant coefficients and its fundamental solutions are $y_1 = e^{-3t}$ and $y_2 = te^{-3t}$. Their Wronskian is $W = \begin{vmatrix} e^{-3t} & te^{-3t} \\ -3e^{-3t} & (1-3t)e^{-3t} \end{vmatrix} = e^{-3t}(1-3t)e^{-3t} - te^{-3t}(-3)e^{-3t} = e^{-6t}$. Substituting this and the right-hand side $g = e^{-3t}/t^2$ into the formula, we get a particular solution $y_p = -e^{-3t}\int \frac{\frac{1}{t^2}e^{-3t}\cdot te^{-3t}}{e^{-6t}}\,\mathrm{d}t + te^{-3t}\int \frac{\frac{1}{t^2}e^{-3t}\cdot e^{-3t}}{e^{-6t}}\,\mathrm{d}t = -e^{-3t}\int \frac{1}{t}\,\mathrm{d}t + te^{-3t}\int \frac{1}{t^2}\,\mathrm{d}t$ $= -e^{-3t}\ln|t| + te^{-3t}(-t^{-1})$. The general solution is then

$$\boxed{y = (-\ln|t| + Bt + C)\,e^{-3t}.}$$

(***b***) **Question** $2t^2y'' + 3ty' - y = t^2\ln t, \quad t > 0\ .$

Solution The homogeneous equation $2t^2y'' + 3ty' - y = 0$ was considered in Q2b, so we know that $y_1 = t^{-1}$, $y_2 = t^{1/2}$. Their Wronskian is $W = \begin{vmatrix} t^{-1} & t^{1/2} \\ -t^{-2} & \frac{1}{2}t^{-1/2} \end{vmatrix} = \frac{3}{2}t^{-3/2}$. Before using the formula for the solution of the nonhomogeneous equation, we should write the equation in the normalised form, which is $y'' + \frac{3}{2}t^{-1}y' - \frac{1}{2}t^{-2}y = \frac{1}{2}\ln t$, so $g = \frac{1}{2}\ln t$. Substituting $g(t)$ and $W(t)$ into the formula, we get a particular solution

$$y_p = -t^{-1}\int \frac{2}{3}t^{3/2}t^{1/2}\frac{1}{2}\ln t\,\mathrm{d}t + t^{1/2}\int \frac{2}{3}t^{3/2}t^{-1}\frac{1}{2}\ln t\,\mathrm{d}t$$

$$= -\frac{1}{3}t^{-1}\int t^2\ln t\,\mathrm{d}t + \frac{1}{3}t^{1/2}\int t^{1/2}\ln t\,\mathrm{d}t$$

Now use integration by parts,

$$\int t^2\ln t\,\mathrm{d}t = \frac{t^3}{3}\ln t - \int \frac{t^3}{3}t^{-1}\,dt = \frac{t^3}{3}\ln t - \frac{1}{9}t^3$$

$$\int t^{1/2}\ln t\,\mathrm{d}t = \frac{t^{3/2}}{3/2}\ln t - \int \frac{t^{3/2}}{3/2}t^{-1}\,dt = \frac{2t^{3/2}}{3}\ln t - \frac{2}{3}\int t^{1/2}\,dt = \frac{2t^{3/2}}{3}\ln t - \frac{2}{3}\frac{t^{3/2}}{(3/2)}\,,$$

ignoring the arbitrary constants of integration. Substituting these integrals back into the expression for y gives

$$y = -\frac{1}{3}t^{-1}\left[t^3\left(\frac{\ln t}{3} - \frac{1}{9}\right)\right] + \frac{1}{3}t^{1/2}\left[t^{3/2}\left(\frac{2}{3}\ln t - \frac{4}{9}\right)\right]$$

which simplifies and ultimately gives the general solution $y = c_1y_1 + c_2y_2 + y_p$ as

$$\boxed{y = \frac{1}{9}t^2(\ln t - 1) + c_1t^{-1} + c_2t^{1/2}.}$$

4. (***a***) **Question**

$$t^2y'' - 3ty' + 4y = 0, \quad t > 0\,.$$

Solution This is an Euler-Cauchy equation. Seeking solutions of the form $y = t^r$ gives $y' = rt^{r-1}$ and $y'' = r(r-1)t^{r-2}$. Substituting into the ODE and canceling the factor of t^r gives the auxiliary equation

$$r^2 - 4r + 4 = 0\,.$$

Thus we have the repeated root $r = 2$, which gives us one solution $y_1 = t^2$.
The other linearly independent solution is found using the method of Reduction of Order, i.e. we write

$$y = t^2v(t)$$

and differentiate (twice) and substitute into the ODE to determine the unknown function $v(t)$. For $t > 0$ you should find that the ODE for $v(t)$ is

$$tv'' + v' = 0\,.$$

Writing $w = v'$ we obtain a separable equation for $w(t)$:

$$tw' + w = 0 \quad \Rightarrow \frac{dw}{w} = -\frac{dt}{t}, \quad t > 0$$

provided that $w \neq 0$ ($w = 0$ is the special solution). Integrating gives the typical solutions

$$\ln|w| = -\ln t + c, \quad \Rightarrow w = At^{-1}, \quad t > 0$$

where $A = \pm e^c$. Allowing $A = 0$ includes the special solution.
Then we obtain $v(t)$ from integrating

$$\frac{dv}{dt} = w = At^{-1} \quad \Rightarrow v = A\int t^{-1}dt + B = A\ln t + B, \quad t > 0$$

and hence the general solution is

$$y = t^2v = t^2(A\ln t + B), \quad t > 0\,.$$

(***b***) **Question**

$$t^2y'' + 3ty' + 5y = 0, \quad t > 0\,.$$

Solution This is an Euler-Cauchy equation. Seeking solutions of the form $y = t^r$ leads to the auxiliary equation

$$r^2 + 2r + 5 = 0\,.$$

Hence we have complex roots $r = -1 \pm 2i = \lambda \pm i\mu$ and the general solution is

$$y = c_1t^{-1}\cos(2\ln t) + c_2t^{-1}\sin(2\ln t), \quad t > 0\,.$$